

Multiple Linear Regression

Predicting one numeric value using many input features.



What Students Will Learn

01

The Problem

Why one feature is not enough

03

Training

Objective function and how the model learns

02

The Formula

MLR equation, coefficients, and predictions

04

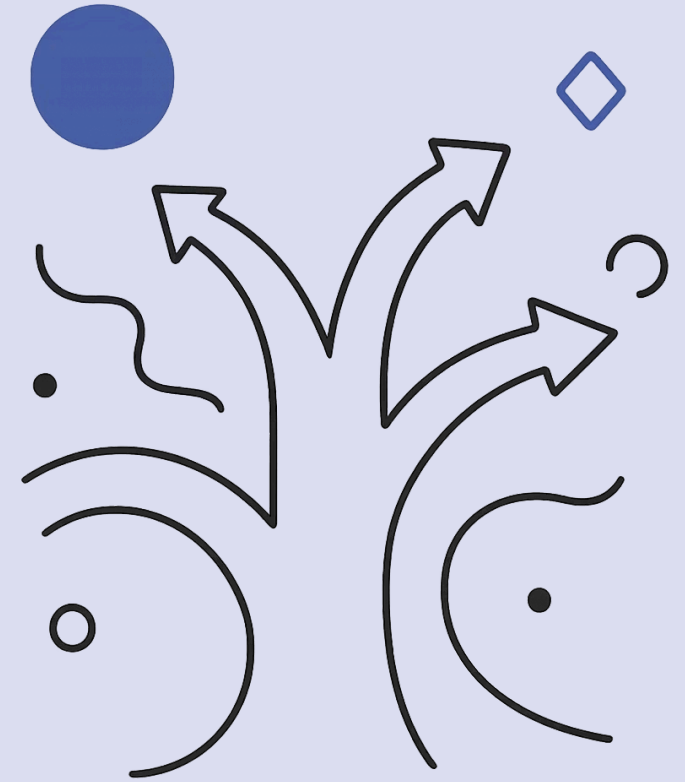
Evaluation

MAE, MSE, RMSE, R^2 , train-test split

PART 1

Foundation

From one input to many inputs.



Simple Linear Regression

Simple Linear Regression uses one feature to make a prediction.

Formula

$$\hat{y} = mx + c$$

Example

$$\text{Predicted Score} = m \times \text{Study Hours} + c$$

Real Life Has Many Reasons

A student's score may depend on many factors — not just one.

Study Hours

Attendance

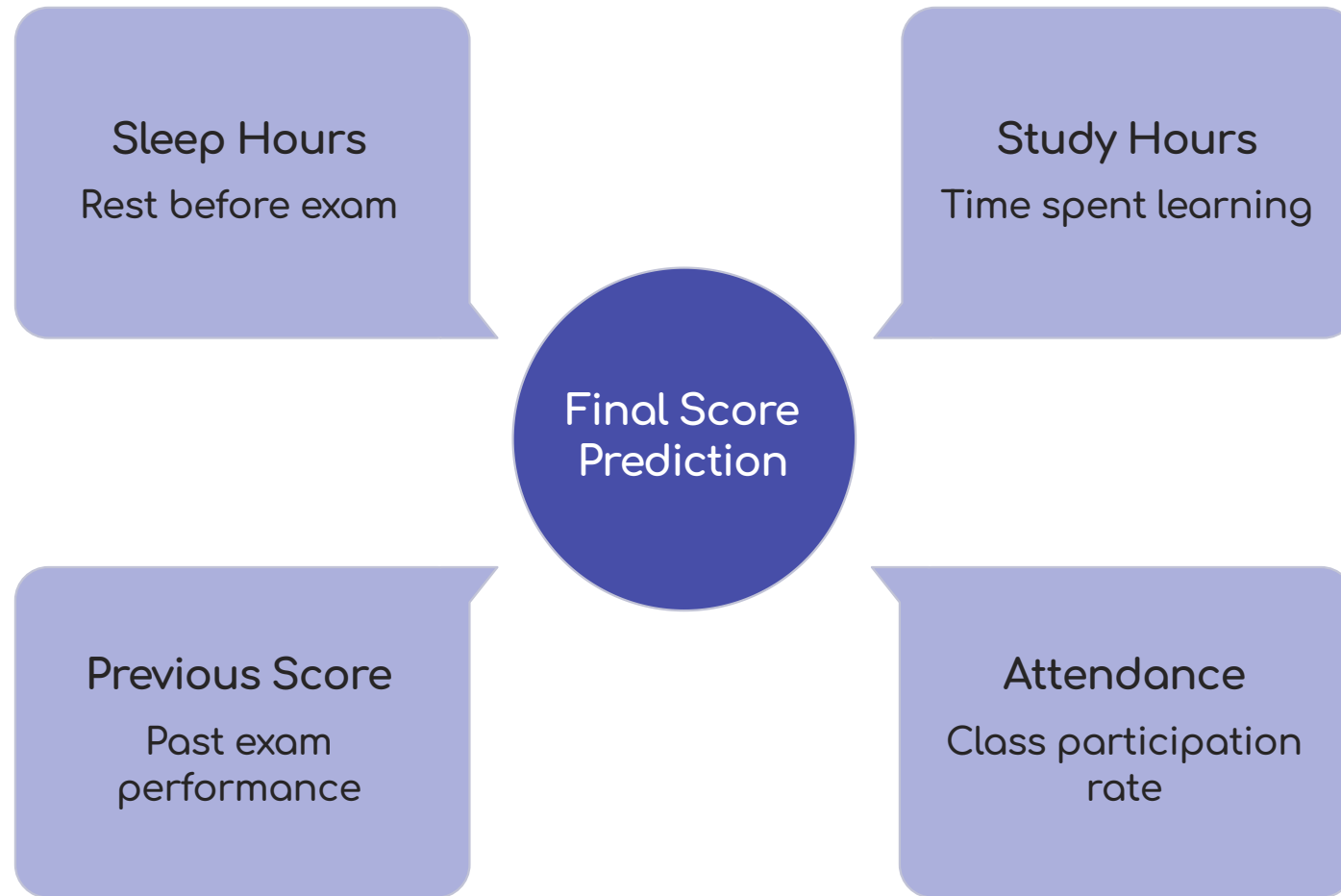
Previous Score

Sleep Hours

Practice Tests

Many Inputs → One Prediction

Multiple Linear Regression predicts one numeric target using multiple features.



Each feature contributes its own share to the final prediction.

Predict Final Exam Score

Question: Can we predict a student's final exam score using academic and lifestyle factors?

Role	Column
Target	Final Score
Features	Study Hours, Attendance, Previous Score, Sleep Hours

Example Dataset

Student	Study Hrs	Attendance	Prev Score	Sleep Hrs	Final Score
A	2	60	40	5	48
B	3	70	45	6	55
C	4	65	50	6	59
D	5	80	55	7	68
E	6	75	60	7	72

Features and Target

Study Hours

Feature

Attendance

Feature

Previous Score

Feature

Sleep Hours

Feature

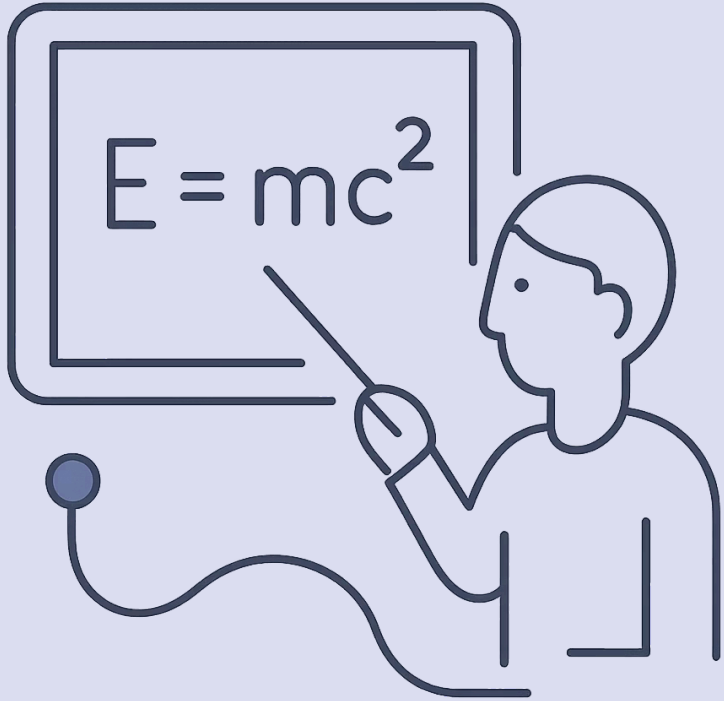
✓ Features are inputs — what we know.

Target is what we want to predict: Final Score.

Simple vs Multiple Regression

Model	Features	Formula
Simple Linear Regression	1	$\hat{y} = mx + c$
Multiple Linear Regression	2 or more	$\hat{y} = b_0 + b_1x_1 + b_2x_2 + \dots$

❏ Same core idea — more features, more coefficients.



PART 2

Formula and Prediction

Understanding the equation clearly.

The Main Formula

$$\hat{y} = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4$$

For student score prediction:

$$\text{Predicted Score} = b_0 + b_1(\text{Study Hrs}) + b_2(\text{Attendance}) + b_3(\text{Prev Score}) + b_4(\text{Sleep Hrs})$$

Formula Symbols

Symbol	Meaning
$\hat{y}$	Predicted value
b_0	Intercept / bias
b_1, b_2	Coefficient for each feature
x_1, x_2	Feature values

❏ Each feature gets its own coefficient.

Different Names for the Same Things

Concept	Other Names
b_0	Intercept, bias, constant
b_1, b_2, b_3	Coefficients, weights, slopes
x_1, x_2, x_3	Features, inputs, independent variables
y	Target, output, dependent variable
$\hat{y}$	Prediction, estimated value

Example Learned Model

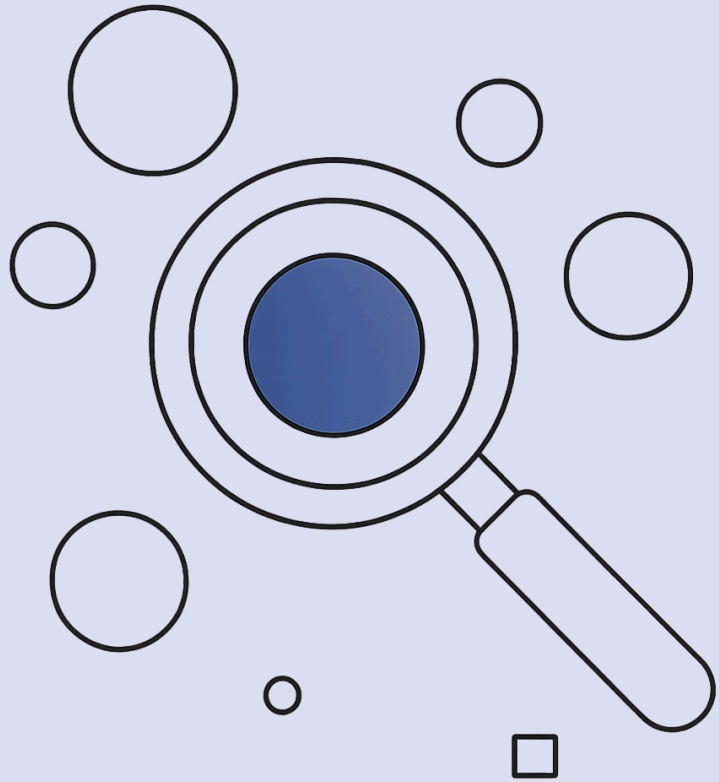
Suppose the model learns these coefficients:

$$\text{Predicted Score} = 5 + 2.5(\text{Study Hrs}) + 0.2(\text{Attendance}) + 0.5(\text{Prev Score}) + 1.5(\text{Sleep Hrs})$$

 This formula combines many feature contributions into a single prediction.

Coefficient Meaning

Feature	Coefficient	Meaning
Study Hours	2.5	1 extra hour adds ~2.5 marks
Attendance	0.2	1 extra point adds ~0.2 marks
Previous Score	0.5	1 extra mark adds ~0.5 marks
Sleep Hours	1.5	1 extra hour adds ~1.5 marks



Keeping Other Features Fixed

Each coefficient means: effect of one feature while all others stay the same.

If Study Hours coefficient = 2.5, then 1 extra study hour adds ~2.5 marks — keeping attendance, previous score, and sleep hours fixed.

Coefficient Sign

Positive Coefficient

Feature increases → prediction increases

Negative Coefficient

Feature increases → prediction decreases

Near Zero Coefficient

Feature has a weak linear effect

❏ Always interpret while keeping other features fixed.

Prediction Example

Model

$$\text{Score} = 5 + 2.5x_1 + 0.2x_2 + 0.5x_3 + 1.5x_4$$

New Student Values

Feature	Value
Study Hours	6
Attendance	80
Previous Score	60
Sleep Hours	7

Prediction Calculation

$$\begin{aligned}\text{Score} &= 5 + (2.5 \times 6) + (0.2 \times 80) + (0.5 \times 60) + (1.5 \times 7) \\ &= 5 + 15 + 16 + 30 + 10.5 = \mathbf{76.5}\end{aligned}$$

✓ Final Score ≈ 76.5

Contribution View

Part	Calculation	Contribution
Base value	5	5
Study Hours	2.5×6	15
Attendance	0.2×80	16
Previous Score	0.5×60	30
Sleep Hours	1.5×7	10.5
Final Prediction	Total	76.5

Why It Is More Realistic

Simple Linear Regression

⚠ Only study hours explain the score.

Multiple Linear Regression

✓ Several factors contribute together — this better matches real-world problems.



House Price Example

$$\text{Price} = b_0 + b_1(\text{Size}) + b_2(\text{Bedrooms}) + b_3(\text{Location}) + b_4(\text{Age})$$

Many features explain one numeric target.



Salary Example

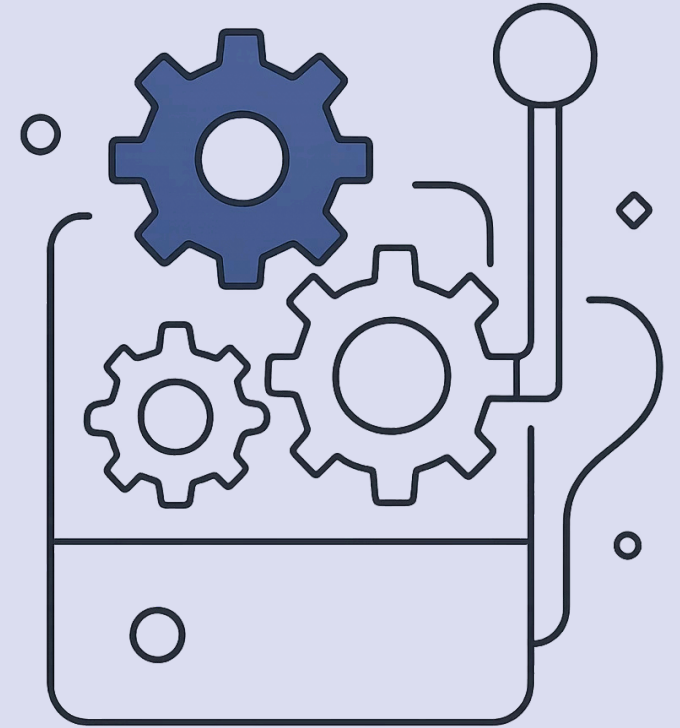
$$\text{Salary} = b_0 + b_1(\text{Experience}) + b_2(\text{Skill Score}) + b_3(\text{Education}) + b_4(\text{Interview Score})$$

Target is numeric. Features are multiple.

PART 3

Training and Evaluation

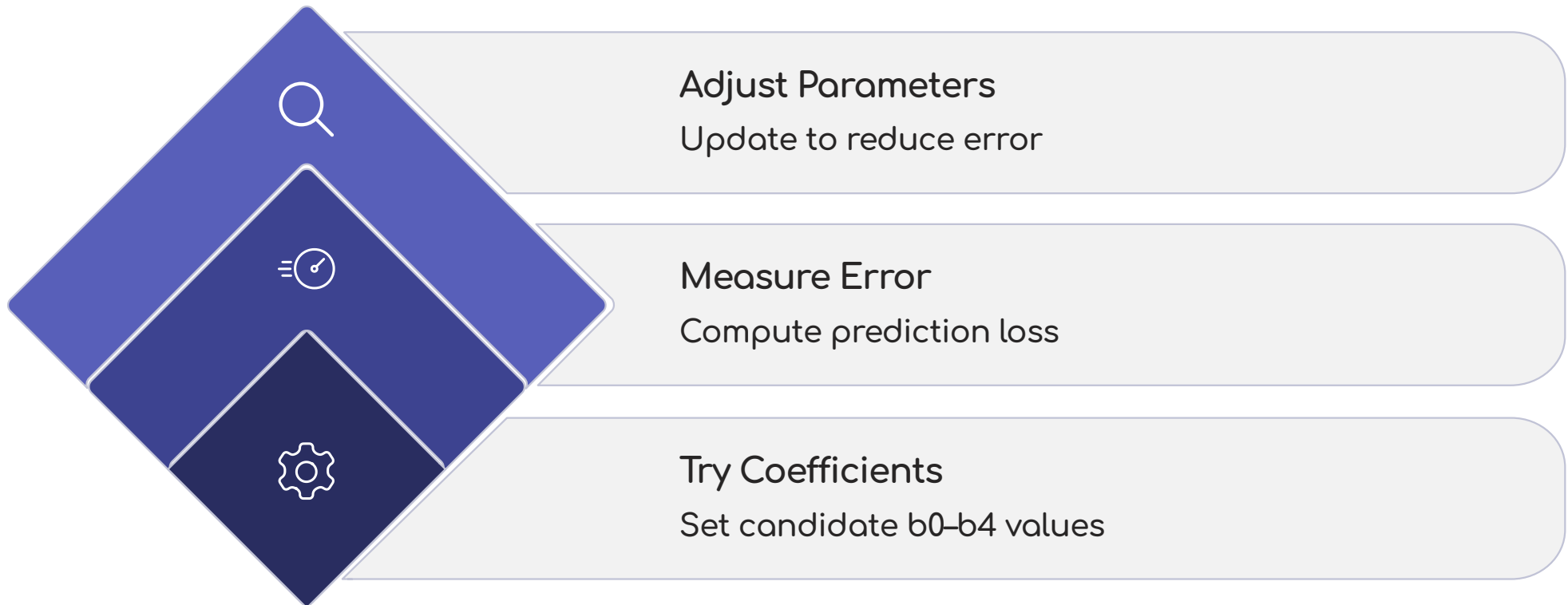
How the model learns and how we judge it.



How the Model Learns

The model tries to find the best values of b_0, b_1, b_2, b_3, b_4 .

 Goal: Make predictions as close as possible to actual values.



Error Formula

$$\text{Error} = y - \hat{y}$$

Actual Score	Predicted Score	Error
70	68	+2
80	83	-3

Squared Error

$$(\text{Squared Error}) = (y - \hat{y})^2$$

Removes Negatives

All errors become positive

No Cancellation

Errors don't cancel each other out

Punishes Large Errors

Bigger mistakes are penalised more

MSE Objective

$$\text{MSE} = \frac{\sum (y - \hat{y})^2}{n}$$

Since $\hat{y} = b_0 + b_1x_1 + b_2x_2 + \dots$, the training goal is:

- ① Find $b_0, b_1, b_2, \dots$ that minimise MSE.

Full Training Goal

$$\text{Minimise } \frac{\sum [y - (b_0 + b_1x_1 + b_2x_2 + b_3x_3 + \dots)]^2}{n}$$

- ✔ Best model = coefficients with the lowest MSE. This is the heart of Multiple Linear Regression.

Coefficients Are Learned Together

The model does not learn each feature separately — it learns all coefficients together.

 Why? Features may be related. Students who study more may also attend more classes — so the model must decide how much credit each feature receives.

Feature Credit Problem

Feature	Possible Overlap
Study Hours	Related to Practice Tests
Attendance	Related to Study Discipline
Previous Score	Related to Current Preparation
Sleep Hours	Related to Concentration

❏ Multiple Regression separates these effects as much as possible.

Matrix Intuition

For many features, we think in table form. A bias column of 1s is added so that $b_0 \times 1 = b_0$.

Bias	Study	Attendance	Previous	Sleep
1	2	60	40	5
1	3	70	45	6
1	4	65	50	6

Coefficient Vector

Coefficient	Meaning
b_0	Intercept
b_1	Study coefficient
b_2	Attendance coefficient
b_3	Previous score coefficient
b_4	Sleep coefficient

 Prediction = feature row $\times$ coefficient vector

Row-Level Prediction

Feature Row

[1, 6, 80, 60, 7]

Coefficient Vector

[5, 2.5, 0.2, 0.5, 1.5]

Calculation

$$1 \times 5 + 6 \times 2.5 + 80 \times 0.2 + 60 \times 0.5 + 7 \times 1.5 =$$

76.5

Actual vs Predicted

Student	Actual	Predicted	Error
A	48	47.6	0.4
B	55	55.2	-0.2
C	59	59.6	-0.6
D	68	67.9	0.1
E	72	72.3	-0.3

✔ Small errors indicate a good fit.

Evaluation Metrics

MAE

Average absolute mistake

MSE

Average squared mistake

RMSE

Error in original unit

R^2

How much variation is explained

Adjusted R^2

R^2 adjusted for number of features

MAE — Mean Absolute Error

$$\text{MAE} = \frac{\sum |y - \hat{y}|}{n}$$

Average absolute prediction error.

 If MAE = 2.5, the model is wrong by around 2.5 marks on average.

MSE — Mean Squared Error

$$\text{MSE} = \frac{\sum (y - \hat{y})^2}{n}$$

Punishes Large Errors

Squaring amplifies bigger mistakes more than smaller ones.

Lower is Better

A lower MSE means predictions are closer to actual values.

RMSE — Root Mean Squared Error

$$\text{RMSE} = \sqrt{\text{MSE}}$$

Error returned to the original unit.

Target in Marks

RMSE is also in marks

Target in Rupees

RMSE is also in rupees

R² Score

R² tells how much variation in the target is explained by the model.

R ² Value	Meaning
1.0	Perfect fit
0.9	Strong fit
0.7	Good fit
0.5	Medium fit
0.0	No better than average prediction

 R² can be negative if the model is very poor.

Adjusted R^2

In Multiple Regression, R^2 can increase when more features are added — even useless ones.



Problem: R^2 may rise with every new feature, even if it adds no real value.



Solution: Adjusted R^2 penalises unnecessary features and rewards useful ones.

R^2 vs Adjusted R^2

Metric	Meaning	Risk
R^2	How much variation the model explains	May increase when more features are added
Adjusted R^2	R^2 adjusted for number of features	Better for comparing multiple regression models

 Use Adjusted R^2 when comparing models with different numbers of features.

Train-Test Split

A model should always be tested on unseen data.

Training Data

Used to learn coefficients

Testing Data

Used to check performance

- Train on old examples. Test on new examples.

Why Testing Matters

If a student practises 10 questions and the exam has the same 10 questions, a high score may not prove understanding.

Similarly, training performance alone is not enough. Testing performance shows whether the model generalises.



Correct ML Workflow

01

Collect Data

03

Split into Train & Test

05

Predict on Test Data

02

Identify Features & Target

04

Train on Training Data

06

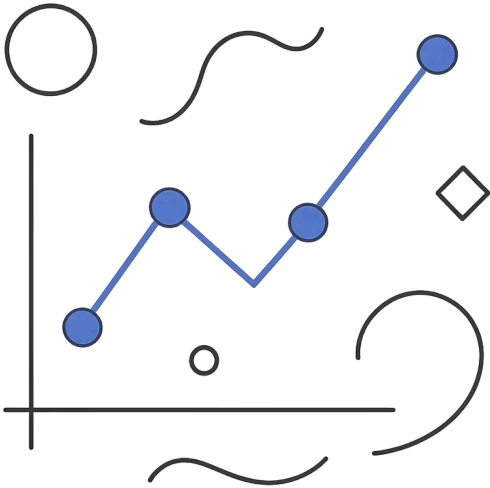
Evaluate & Improve

Training vs Testing Error

Case	Training Error	Testing Error	Meaning
Underfitting	High	High	Model too simple
Good Fit	Low	Low	Model generalises well
Overfitting	Very Low	High	Model memorised training data

✓ Goal: Low testing error.

Underfitting



The model is too simple and fails to learn the real pattern.

⚠ Example: Predicted Score = 60 for every student.

Result: High training error *and* high testing error.



Overfitting

The model memorises training data too much.

Training Error

Very low — fits training data perfectly

Testing Error

High — fails on new, unseen data



Overfitting means the model does not generalise.